



The company plans to leverage Quantron's expertise in designing retro-fitment kits and develop similar kits for the Indian market, tailored to Indian pricing, licensing, and approvals. Vikram Sinha, Head of Manufacturing at ETO Motors, explains that retrofitting trucks with EV powertrains at the end of their

conventional truck lifecycle can save truck owners from major repairs and expenses.

In conclusion, ETO Motors is a leading innovator in the three-wheeler EV manufacturing industry, driving inclusivity and sustainability. By actively employing women drivers and producing cutting-edge

electric vehicles, they are revolutionising both the automotive industry and societal norms. As we strive for a more sustainable future, ETO Motors serves as an inspiring example, showcasing the possibilities that arise when diversity and innovation converge.

—Sharad Bhowmick

2 ELECTRIFYING THE PEN-PAPER CIRCUITRY

This 2021-born Kanpur based startup, called Likhtronics, is trying to make circuit wires redundant with their eco-friendly inks, and is working to revolutionise PCB fabrication with an ink that can be screen printed

Did you ever get entangled in wires while demonstrating an electric circuit? What if you could just draw the circuit with a pen on paper and the lines you drew could act as wires? National Center for Flexible Electronics (NC-FlexE) has developed such a formulation after five years of efforts and put it into an ink and pen format. The IIT Kanpur based startup, called Likhtronics, has commercialised it.

The startup's inks are proprietary formulations to be used by the manufacturers of PCBs, membrane switches, sensor electrodes, and remote controls, etc. The startup also offers educational kits based on these specialised inks, which can be used for training basic electronics to students of all age groups.

"While imported specialised inks from Asia (China, Taiwan, etc) have several issues, including quality, imported inks from the West are expensive," says the Founder

and Managing Director of Likhtronics, Dr K. Sudheendra Rao, who first sensed the problem when he was associated with a company that was developing biosensors. He felt the need to develop an indigenous ink ecosystem in the country. Most of their inks are now screen printable; their formulation for inkjet printing is under development.

Though not erasable, Likho-

tronics inks remain stable at room temperature, have a longer shelf-life than the normal industry standard of six months, and are long-lasting on paper, if maintained properly, claims Dr Rao. They customise the inks based on the substrate it is to be used on.

"The industry can use these inks and develop various printable and flexible electronic products like RFID tags, solar panels, and PCBs for various applications," claims Dr Rao. They claim use of mostly eco-friendly materials and assert, "Our pens are as safe as your regular pens."

Dr Rao points out that, currently, PCB production uses a chemical etching process, which is hazardous. Likhtronics claims to have developed an ink that can be screen printed, and hence is eco-friendly. Dr Rao says, "We are very close to it and our customers have validated the process. But still, it requires a lot

